

Question	Marking Guidance	Mark	Comments
08.1	C	1	Auto mark
08.2	1. No separation of chromatids/chromosomes/centromeres; 2. Chromatids/chromosomes all go to one pole/end/sides of cell/not pulled to opposite poles; 3. Doubles chromosome number in cell/one daughter cell gets no chromosomes or chromatids;	2 max	1. Accept anaphase prevented 1. Accept nondisjunction 1.Reject homologous pairs 3. Accept DNA for chromosomes 3. Accept ploidy 3. Ignore references to 'genetic information' Ignore simple descriptions of what normally happens in mitosis
08.3	1. (No, because) at 100 there are still some (7%) cancer cells dividing/undergoing mitosis; 2. So, cancer not destroyed/may continue to grow/spread/form tumours; 3. Best concentration may be between 100 and 1000/need trials between 100 and 1000; 4. This research in culture, don't know effect of KI on people; 5. (Yes, because) above 100 produces little increase in % of cells not dividing/undergoing mitosis/at 100, most (93%) cancer cells unable to divide/dead; 6. Above 100 may be harmful (to body); 7. Higher concentrations more expensive; 8. (above 100) will have more effect on (rapidly dividing) cancer cells;	4 max	1. Accept idea that all division stops only at 1000 2. Must refer to cancer spreading not cells dividing 4. Reject 'not tested on humans' 4.Reject 'done in animals' 5. Must clearly link lack of monopolar mitotic spindles with cell division 6. Accept 'above 100/high concentrations produce harmful side effects/named effects' 8.Must relate to 100

08.4	1. 10 cm³ of 10 000 nmol dm⁻³/ (original) solution; 2. 90 cm³ of water;	2	If ratio correct but make wrong volume e.g. 1 litre, award 1 mark
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